

Practical no. 17

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Class :- CO3K – B

Subject :- OOP

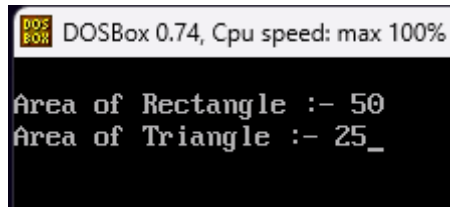
Aim :- Write programs for :- 1. Pointer to derived class in single inheritance
2. Pointer to derived class in Multilevel Inheritance

Program :-

```
#include<iostream.h>
#include<conio.h>
class Polygon{
public:
int width,height;
void set_values(int w,int h){
width=w;
height=h;
}
virtual float area(){
return 0;
}
};
class Rectangle:public Polygon{
public:
float area(){
return width*height;
}
};
class Triangle:public Polygon{
public:
float area(){
return (width*height)/2.0;
}
};
void main(){
Rectangle r;
Triangle t;
```

```
Polygon* p;  
clrscr();  
p=&r;  
p->set_values(10,5);  
cout<<"\nArea of Rectangle :- "<<p->area();  
p=&t;  
p->set_values(10,5);  
cout<<"\nArea of Triangle :- "<<p->area();  
getch();  
}
```

Output :-

A screenshot of a DOSBox 0.74 window. The title bar at the top reads "DOSBox 0.74, Cpu speed: max 100%". The main window area has a black background with white text. The text displayed is "Area of Rectangle :- 50" on the first line and "Area of Triangle :- 25_" on the second line. The text is in a monospaced font, typical of a terminal or DOS window.

```
DOSBox 0.74, Cpu speed: max 100%  
Area of Rectangle :- 50  
Area of Triangle :- 25_
```